

Computer Forensics Exam

Full Marks: 70

Section A

Answer all the MCQs

(1x10=10)

1. Which of the following is a key principle in digital forensics?

- a) Preservation of evidence
- b) Data destruction
- c) Data encryption
- d) Data obfuscation

2. What is the primary difference between volatile and non-volatile data in digital forensics?

- a) Volatile data can be easily recovered, while non-volatile data cannot.
- b) Volatile data is stored on cloud servers, while non-volatile data is stored on local devices.
- c) Volatile data is temporary and lost when power is cut, while non-volatile data persists even without power.
- d) Volatile data is easier to encrypt than non-volatile data.

3. Which of the following is an example of an open-source digital forensics tool?

- a) Oxygen Forensic Detective
- b) EnCase

- c) Autopsy
- d) Cellebrite

4. In a digital forensic context, what is the primary purpose of metadata?

- a) To store user-generated data
- b) To provide a timestamp for files
- c) To maintain the chain of custody
- d) To recover deleted files

5. What does the term "hash value" refer to in digital forensics?

- a) A cryptographic key used for encryption
- b) A unique identifier for a file or data set
- c) A type of malware
- d) A digital signature for authentication

6. In the context of digital forensics, what does the acronym "RAM" stand for?

- a) Random Access Memory
- b) Read-Only Memory
- c) Redundant Array of Memory
- d) Remote Access Module

7. Which phase of the digital forensics process involves the examination and analysis of collected digital evidence?

- a) Acquisition
- b) Preservation
- c) Examination
- d) Reporting

8. What does the term "chain of custody" refer to in digital forensics?

- a) A type of encryption technique
- b) The chronological documentation of evidence handling
- c) The process of recovering deleted files
- d) A forensic analysis tool

9. Which of the following is a common challenge in mobile device forensics?

- a) Lack of encryption on mobile devices
- b) Limited storage capacity of mobile devices
- c) A wide range of operating systems and device models
- d) Mobile devices are not susceptible to malware

10. What is the primary goal of steganography in the context of digital forensics?

- a) To encrypt data

- b) To recover deleted files
- c) To hide data within other data to avoid detection
- d) To create forensic reports

Section B

1. **The SOC alerted Janice from Accounting that her corporate credentials were discovered on a dark web marketplace. Threat intelligence linked the exposure to suspicious activity originating from her workstation. The IR team recovered several potentially malicious files from her system and forwarded them to the malware analysis team. The files are suspected to be part of an information-stealing malware campaign.**

Download the files:

<https://drive.google.com/file/d/1bivek7AdkFRoBG1MgNTsNthctmMz26Zn/view?usp=sharing>

Password- infected

You are part of the malware analysis team. Analyze the recovered malware and answer the following questions with proper screenshots: (20)

- a) What is the SHA-256 hash of the malware binary?
- b) Which programming language and version was used to develop the malware?
- c) Several GitHub repositories are referenced in the static strings. Which repository most likely indicates the malware's data exfiltration capability?
- d) Which GitHub repository, used as a dependency, supports Janice's belief that she had downloaded a harmless screenshot utility?
- e) Which specific function call in the code suggests that the malware creates a file upon execution?
- f) Which file is repeatedly written to disk by the malware?
- g) Janice's new password was captured and saved to a file. What are her username and password as recorded?
- h) Which application was open on Janice's system when she last executed the so-called "screenshot app"?
- i) The malware exfiltrates data using FTP. What is the domain to which it sends the stolen data?
- j) What credentials are being used by the threat actor for accessing the FTP exfiltration server?

2. **A Windows Endpoint was recently compromised. You were given the memory dump of the compromised host. You should continue to investigate. Perform memory forensics on the memory dump and answer the following questions. (20)**

- a) What was the date and time when Memory from the compromised endpoint was acquired?
- b) What was the suspicious process running on the system?
- c) Analyze and find the malicious tool running on the system by the attacker.
- d) What is the compromised user password?

Attach the screenshots for every answer.

Download the memory dump from this link:

https://drive.google.com/file/d/12vkQFSgZQkTu89g7NlIZsqW80SCP8O9_/view?usp=sharing

Password:- infected

3. **Answer the following questions by analyzing this email:**

<https://drive.google.com/file/d/1BuSsvzAh2BxBgm1ptnKPezyh1bKwsp3y/view?usp=sharing> (10)

- 1) What is the sender's IP address? _____
- 2) The mail is genuine: _____ (True/False)
- 3) What is the SPF result for this mail? _____
- 4) Which mail server is used to send the mail? _____
- 5) If you will reply to this mail, to which email address your mail will be sent to? _____

4. **Answer the following questions by analyzing this packet capture file:**

https://drive.google.com/file/d/1SD2vnPZs6_EbK0g9b1y9oKdhaOQfSlZo/view?usp=drive_link (10)

- 1) Which attack was performed? _____
- 2) What was the attacker's IP? _____
- 3) What was the victim's IP? _____
- 4) Attack was successful or not? _____
- 5) If the attack was successful, what are the credentials? _____

FINAL EXAM

12th May 2025



Exam Answer Sheet

Name : Sarthak Das

Session : 11th February – 22nd April 2025

For the class of Computer Forensic, taught by

Drabanti Boral

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1 Section A: MCQs

1.1 Question 1

- (a) Preservation of evidence

1.2 Question 2

- (c) Volatile data is temporary and lost when power is cut, while non-volatile data persists even without power.

1.3 Question 3

- (c) Autopsy

1.4 Question 4

- (b) To provide a timestamp for files

1.5 Question 5

- (b) A unique identifier for a file or data set

1.6 Question 6

- (a) Random Access Memory

1.7 Question 7

- (c) Examination

1.8 Question 8

- (b) The chronological documentation of evidence handling

1.9 Question 9

- (c) A wide range of operating systems and device models

1.10 Question 10

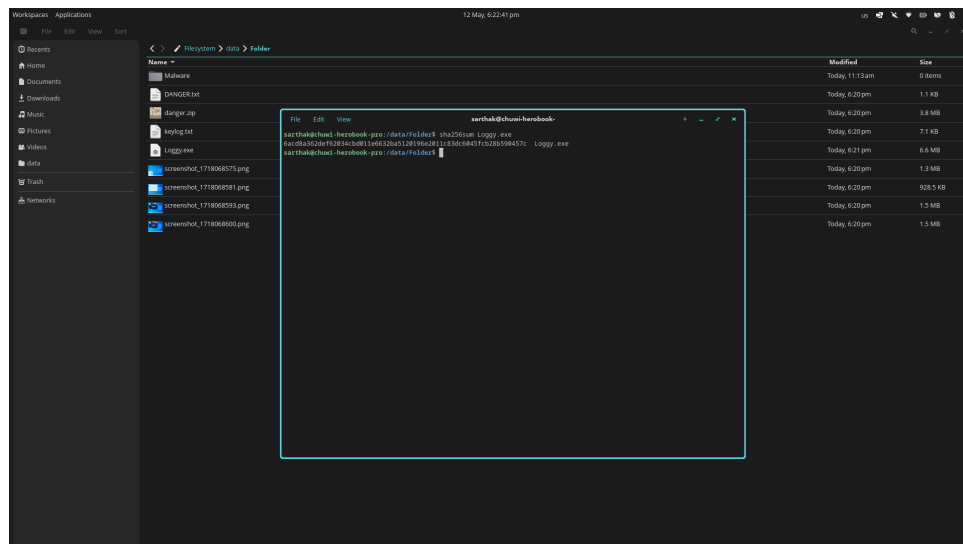
- (c) To hide data within other data to avoid detection

2 Section B: Practical

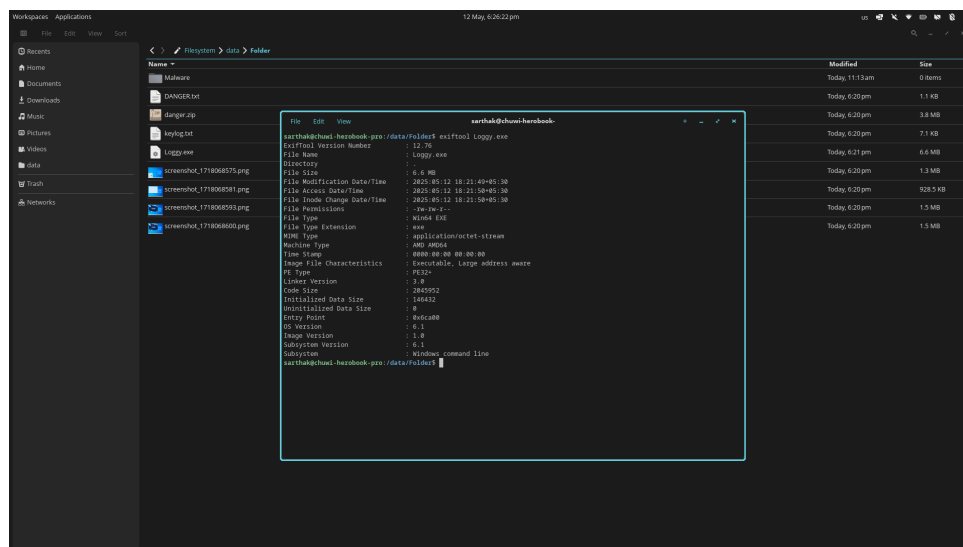
2.1 Question 1

We analyze the malware in a Linux (Pop!_OS 24.04 LTS) system.

Part a. 6acd8a362def62034cbd011e6632ba5120196e2011c83dc6045fcb28b590457c



Part b. This is a Windows command line executable (PE32+) with OS version 6.1 and Linker version 3.0.



Part c. github.com/jlaffaye/ftp[illegible]

Part d. github.com/kbinani/screenshot

[illegible]

Part e. syscall.CreateFile

[illegible]

Part f. keylog.txt

[illegible]

Part g. USERNAME: Janice, PASSWORD: Password123...

[illegible]

Part h. Notepad, or any other text editor as indicated by the previous screenshot (reference to ".txt").

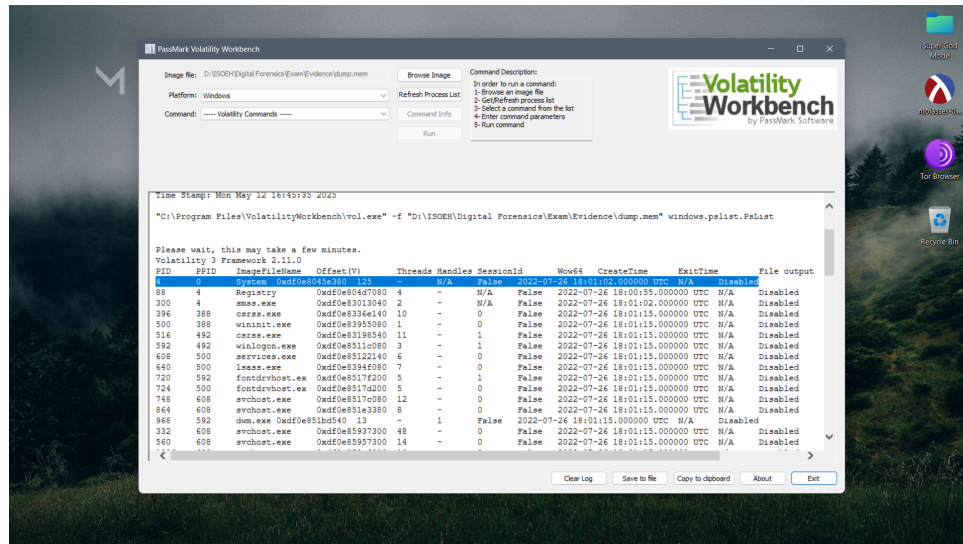
Part i. orange.com

[illegible]

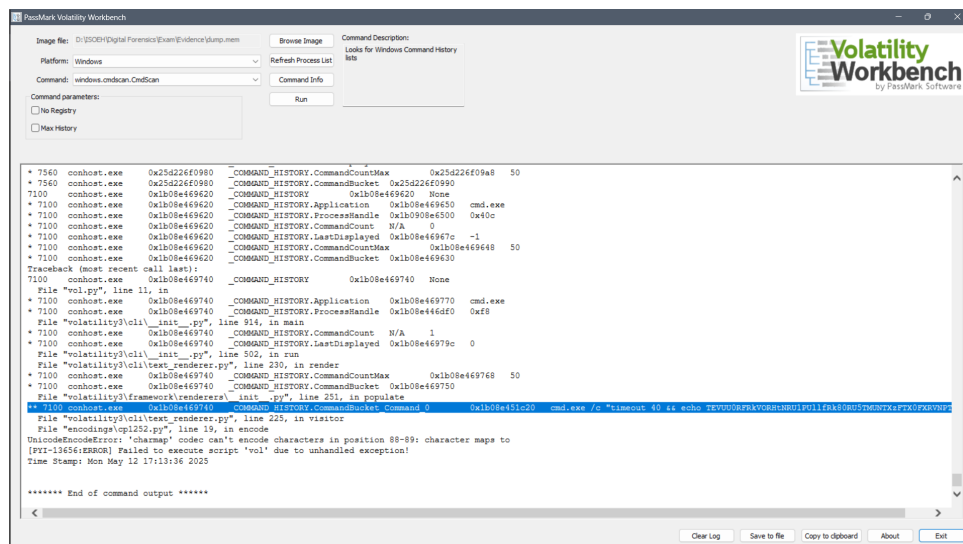
2.2 Question 2

We use Volatility Workbench V3.0 Build 1010 to analyze this memory dump file.

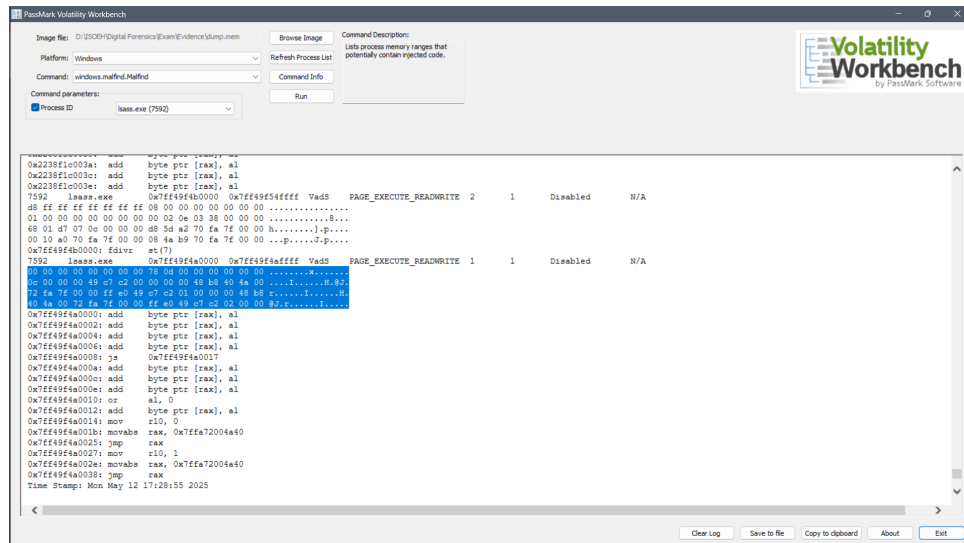
Part a. On running a `pslist` command, we find July 26, 2022 at 18:01:02 UTC, which is the CreateTime for the process `System`.



Part b. On running a `cmdscan` command, we find `conhost.exe` with PID 7100 hosting a `cmd.exe` shell, which seems suspicious.

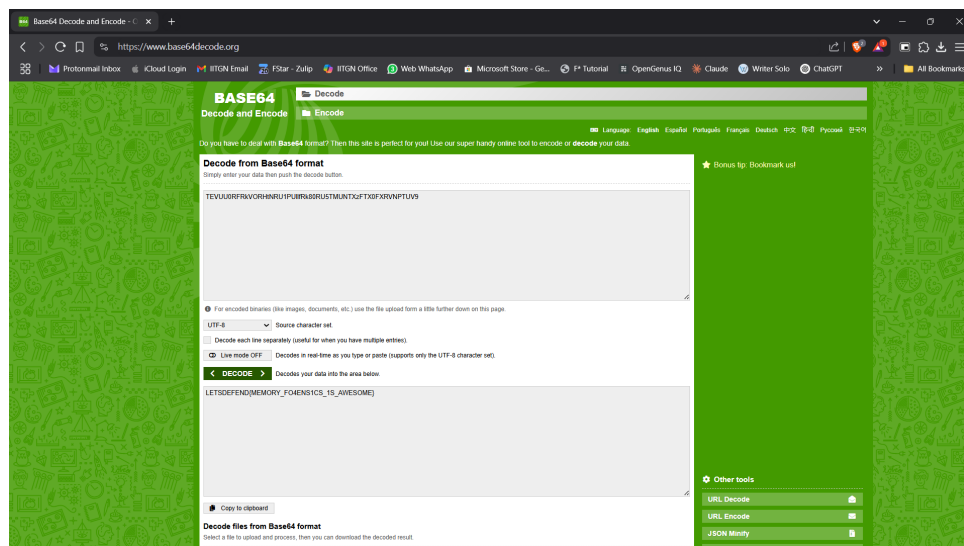


Part c. On running a `malfind` command on the `lsass.exe` (PID 7592), which was the LSASS handle for the suspicious `conhost.exe` process, we find the following injected code.



This is a memory dump, when taken into account along with an existing `ruby.exe` (PID 1908) running with 9 threads, hints as a potential Metasploit attack – most likely a `get_credentials` post module via kiwi (Mimikatz).

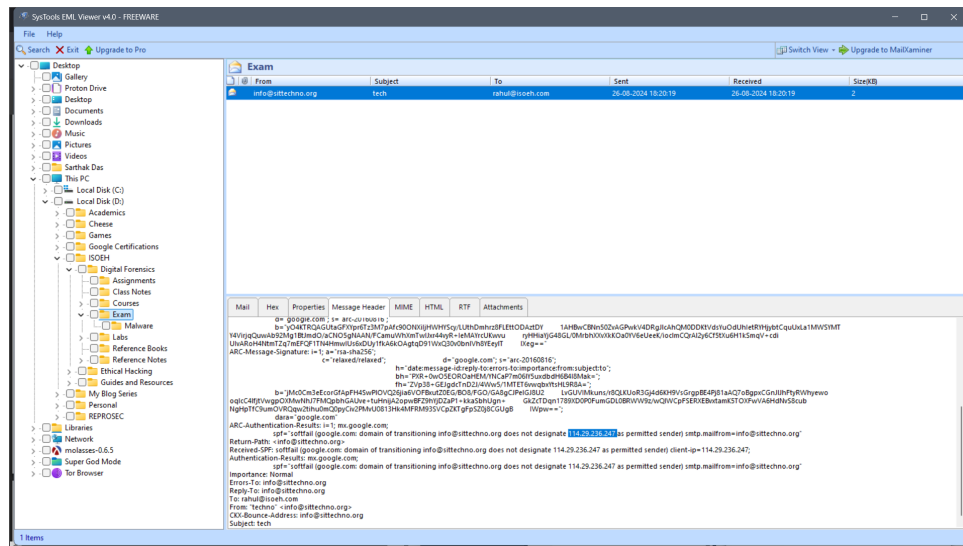
Part d. Decoding the Base64 blob we get from the `cmdscan`, we get `LETSDEFEND{MEMORY_F04ENS1CS_1S_AWESOME}`, which seems to be the compromised password.



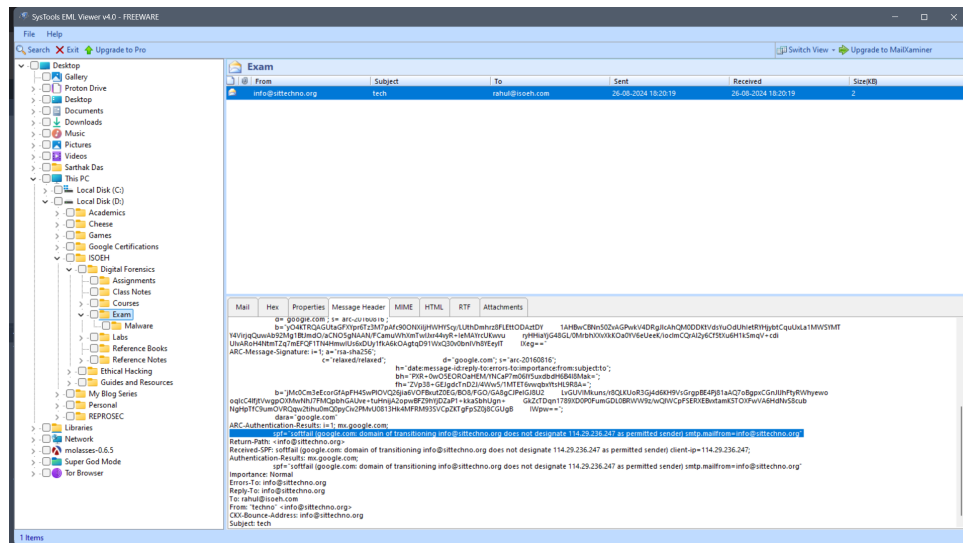
2.3 Question 3

We use SysTools EML Viewer v4.0.0.0 to analyze this email file.

Part 1. 114.29.236.247

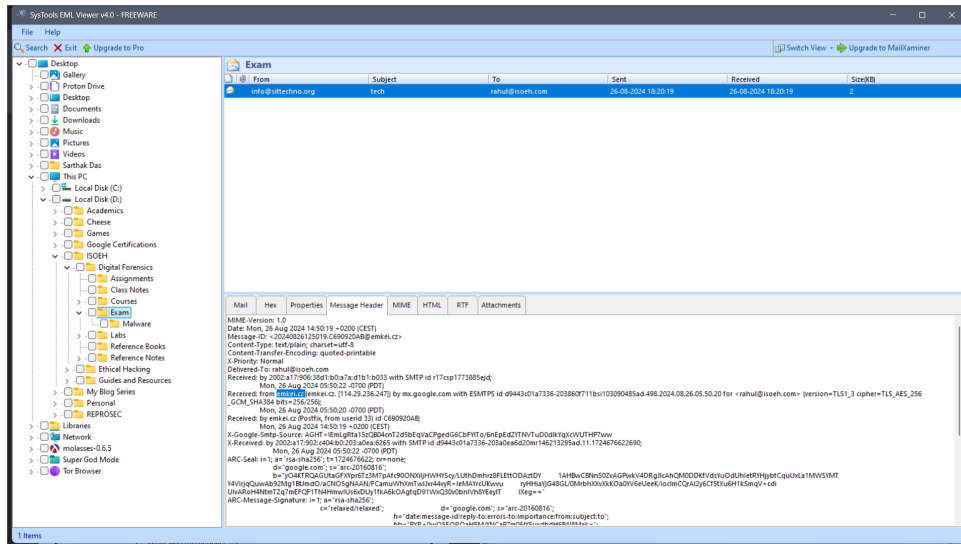


Part 2. False.

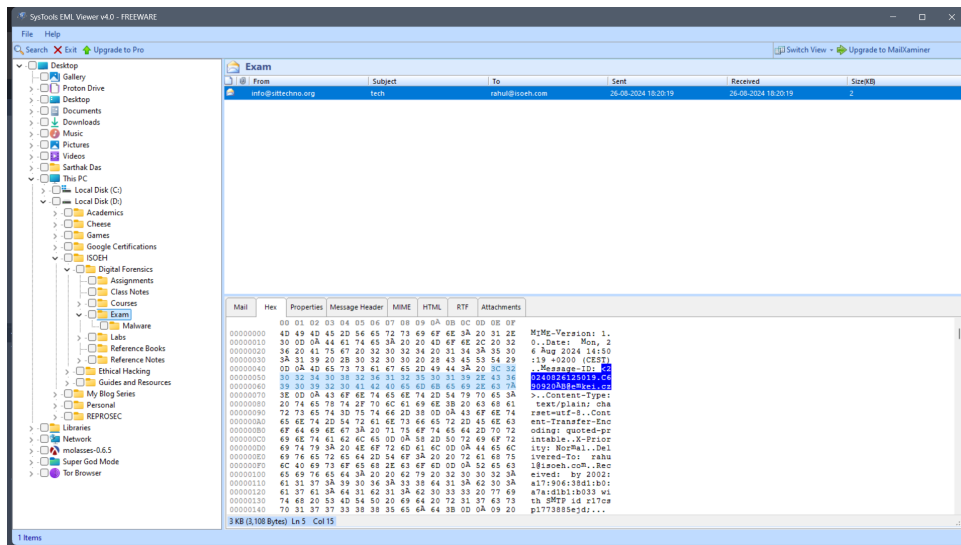


Part 3. “softfail (google.com: domain of transitioning info@sittechno.org does not designate 114.29.236.247 as permitted sender) smtp.mailfrom = info@sittechno.org”, as shown in the previous screenshot.

Part 4. emkei.cz



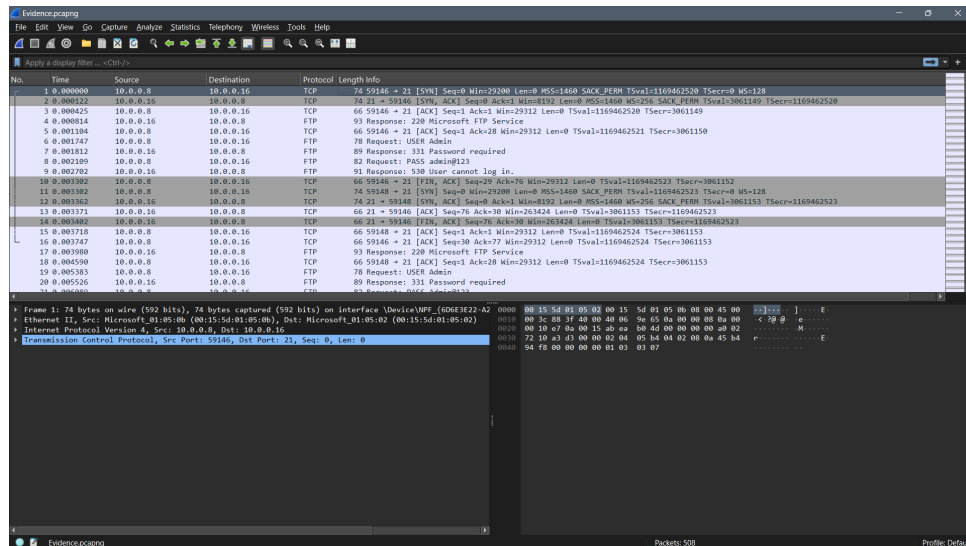
Part 5. 20240826125019.C690920AB@emkei.cz



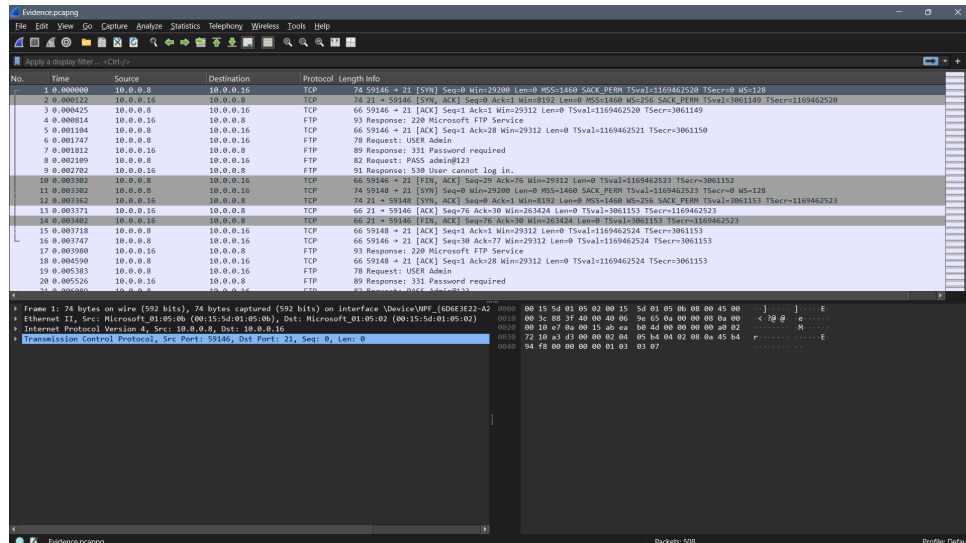
2.4 Question 4

We use Wireshark Version 4.4.6 to analyze this packet capture file.

Part 1. This is an FTP brute-force dictionary attack.

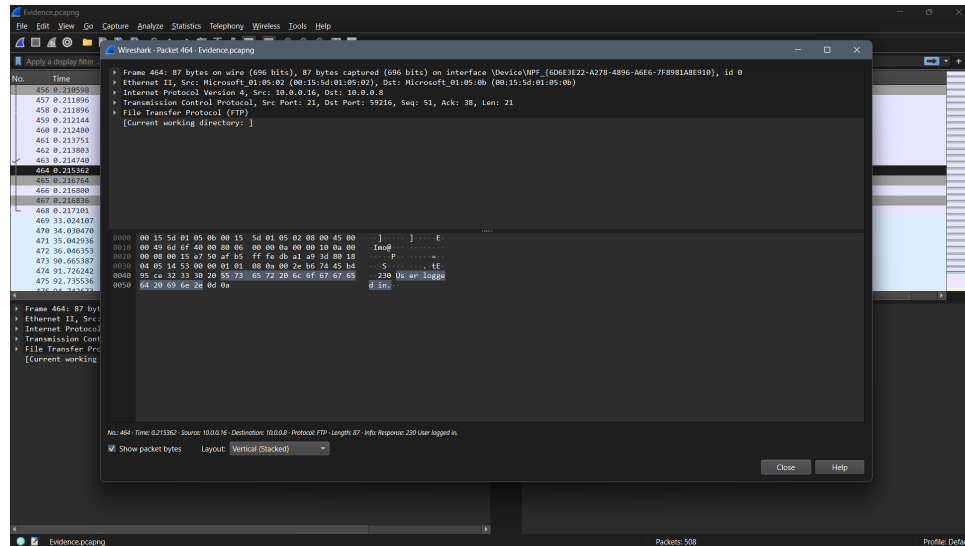


Part 2. 10.0.0.8



Part 3. 10.0.0.16, as shown in the previous screenshot.

Part 4. The attack was successful.



Part 5. User: Administrator, Password: qwerty@123

